

Nationwide House Energy Rating Scheme®

NatHERS® Certificate No. #HR-GQMKIV-01

Generated on 25 Jun 2026 using Hero 4.1 (Chenath v3.23)

Property

Address 18 Ulm Street, Ermington, NSW, 2115
Lot/DP 32/-/DP31884
NCC Class* 1a
Floor/all Floors 1 of 2 floors
Type New

Plans

Main Plan REV I - 23/06/2026
Prepared by ENES DESIGN & CONSTRUCTION PTY LTD

Construction and environment

Assessed floor area (m²)*		Exposure Type
Conditioned*	251.0	Suburban
Unconditioned*	2.4	NatHERS climate zone
Total	291.8	56 - Mascot AMO
Garage	38.3	



Accredited assessor

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Accreditation No. 10205
Assessor Accrediting Organisation HERA
Declaration of interest No Conflict of Interest

NCC Requirements

BCA provisions Volume 2
State/Territory variation Yes

National Construction Code (NCC) requirements

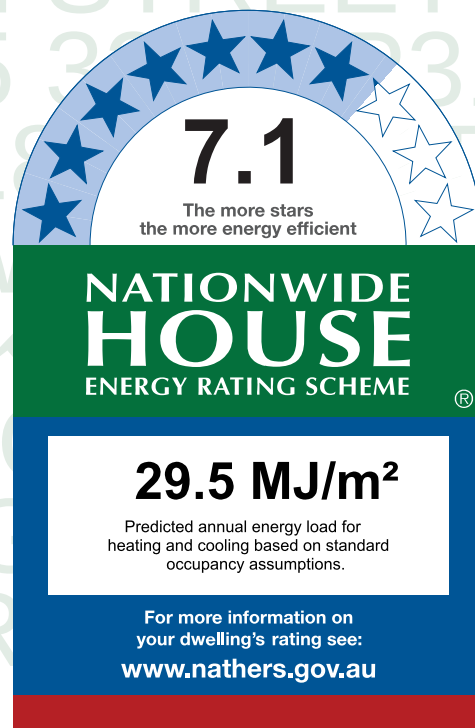
The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J2D2(2)(a) and (3) of NCC Volume One.

NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.

The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au.

Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.

Thermal performance star rating



Thermal performance (MJ/m²)

Limits taken from ABCB Standard 2022

	Heating	Cooling
Modelled	19.1	10.4
Load limits	25	18

Features determining load limits

Floor type
(lowest conditioned area) CSOG
NCC climate zone 1 or 2 N
Outdoor living area N
Outdoor living area ceiling fan N

Whole of Home performance rating

No Whole of Home performance rating generated for this certificate.

Verification

To verify this certificate, scan the QR code or visit
<http://www.hero-software.com.au/pdf/HR-GQMKIV-01>.

When using either link, ensure you are visiting <http://www.hero-software.com.au>



* Refer to glossary.

About the ratings

Thermal performance rating

NatHERS thermal software models the expected heating and cooling energy loads using information about the design, construction, climate and common patterns of household use. The thermal performance rating (shown as a star rating on this Certificate) does not take into account appliances, apart from the airflow impacts from ceiling fans.

Whole of Home performance rating

NatHERS Whole of Home software uses the heating and cooling energy loads combined with the energy performance of the home's appliances (heating, cooling, hot water, lighting, pool/spa pump and onsite renewable energy generation and storage) and models the expected energy value* of the whole home. The Whole of Home performance rating is shown as a score out of 100 on this Certificate.

Heating and Cooling Load Limits

Additional information

In some locations under the NCC NatHERS pathway, separate heating and cooling load limits may apply. Minimum required star ratings in northern parts of Australia may also be affected by the presence or absence of an outdoor living area and/or an outdoor living area ceiling fan. Refer to the *ABCB Standard: NatHERS heating and cooling load limits* for details or contact the relevant local building regulating authority, noting that State and Territory variations may also apply.

Setting options:

Floor type:

- CSOG - Concrete Slab on Ground
- SF - Suspended Floor (or a mixture of CSOG and SF)
- NA - Not Applicable

NCC climate Zone 1 or 2:

- Yes
- No
- NA - Not Applicable

Outdoor living area:

- Yes
- No
- NA - Not Applicable

Outdoor living area ceiling fan:

- Yes
- No
- NA - Not Applicable



Predicted onsite renewable energy impact

No Whole of Home performance assessment conducted for this certificate.

Predicted Whole of Home annual impact by appliance

Shows the contribution each appliance has on the home's annual energy use, greenhouse gas emissions and cost without solar.

Energy use:

No Whole of Home performance assessment conducted for this certificate.

Greenhouse gas emissions:

No Whole of Home performance assessment conducted for this certificate.

Cost:

No Whole of Home performance assessment conducted for this certificate.

Certificate check

The checklist covers important items impacting the dwelling's ratings.

It is recommended that the accuracy of the whole certificate is checked.

Note: The boxes indicate when and who should check each item.

It is not mandatory to complete this checklist.

Approval stage		Construction stage		
Assessor checked	Consent authority/ surveyor checked	Builder checked	Consent authority/ surveyor checked	Occupancy/other

Genuine certificate check

Does this Certificate match the one available at the web address or QR code verification link on the front page?

☐☐☐☐

Does the NatHERS certificate number on the NatHERS-stamped plans match the number on this Certificate?

☐☐☐☐

Thermal performance check

Windows and glazed doors

Does the window size, opening type and location shown on the NatHERS-stamped plans or as installed match what is shown in 'Window and glazed door schedule' and 'Roof window schedule' tables on this Certificate?

☐☐☐☐☐

Does the installed windows meet the substitution tolerances (AFRC* based SHGC* and U-values*) as shown in the 'Window and glazed door type and performance' and 'Roof window type and performance' tables on this Certificate?

☐☐☐

External walls

Does the external wall bulk insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'External wall type table' on this Certificate?

☐☐☐☐☐

Does the external wall shade (colour) match what is shown in the 'External wall type' table on this Certificate?

☐☐☐☐☐

Floor

Does the floor insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Floor type' table on this certificate?

☐☐☐☐☐

Ceiling penetrations*

Does the 'quantity' and 'type' of ceiling penetrations* (e.g. downlights, exhaust fans, etc) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling penetrations' table on this Certificate?

☐☐☐☐☐

Ceiling

Does the ceiling insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling type' table on this Certificate?

☐☐☐☐☐

Roof

Does the external roof shade (colour) on the NatHERS stamped plans or as installed match what is shown in the 'Roof type' table on this Certificate?

☐☐☐☐☐

Apartment entrance doors (NCC Class 2 assessments only)

Does the 'External Door Schedule' show apartment entrance doors? Please note that an "external door" between the modelled dwelling and a shared space, such as an enclosed corridor or foyer, should not be included in the assessment (because it overstates the possible ventilation) and would invalidate the Certificate.

☐☐☐☐

Exposure*

Has the appropriate exposure type (terrain) (shown on page 1) been applied? For example, it is unlikely that a ground-floor apartment is "exposed" or a top floor high-rise apartment is "protected".

☐☐☐☐

Heating and cooling load limits*

Do the load limits settings (shown on page 1) match what is shown on the NatHERS-stamped plans?

☐☐☐☐☐

* Refer to glossary.

Certificate check

Continued

Approval stage		Construction stage		
Assessor checked	Consent authority/ surveyor checked	Builder checked	Consent authority/ surveyor checked	Occupancy/other

Additional NCC requirements for thermal performance (not included in the NatHERS assessment)

Thermal bridging

Does the dwelling meet the NCC requirement for thermal bridging?		☐	☐	☐	☐
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Insulation installation method

Has the insulation been installed according to the NCC requirements?			☐	☐	☐
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Building sealing

Does the dwelling meet the NCC requirements for Building Sealing?		☐	☐	☐	☐
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Whole of Home performance check (not applicable if a Whole of Home assessment is not conducted)

Appliances

Does the cooling appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the heating appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or as installed, match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the hot water system type and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the pool pump efficiency/performance shown on the NatHERS-stamped plans or as installed match the minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the onsite renewable energy system type, orientation and system size or generation capacity shown on the NatHERS stamped plans or installed match the 'Onsite Renewable Energy schedule' on this Certificate?	☐	☐	☐	☐	☐

Additional NCC Requirements for Services (not included in the NatHERS assessment)

Does the lighting meet the artificial lighting requirements specified in the NCC?		☐	☐	☐	☐
Does the hot water system meet the additional requirements specified in the NCC?		☐	☐	☐	☐

Provisional values* check

Have provisional values* been used in the assessment and, if so, are they noted in 'Additional notes' table below?	☐	☐			
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Other NCC requirements

Note: This Certificate only covers the energy efficiency requirements in the NCC. Additional requirements that must also be satisfied include, but are not limited to: condensation, structural and fire safety requirements and any state or territory variations to the NCC energy efficiency requirements.

Additional Notes

- Shading to existing fence line and house modelled as per NatHERS technical Notes
- The side and rear fence heights of the neighbouring building are to be 1.8m where local planning requirements are unknown.
- Heights of the neighbouring building have been modelled to include all known site level changes that will impact on shading the dwelling being rated.

FLOORING:

- Slab on ground, waffle pod construction - NIL extra insulation required
- Suspended particleboard/timber flooring - NIL insulation required

WALLS:

- Brick Veneer/Cladding with R2.5 bulk insulation to external walls EXCLUDING garage external walls
- Garage external walls to have NIL bulk insulation
- Garage internal partition walls to have R2.5 bulk insulation

CEILING & ROOF:

- R4.0 insulation at ceiling level
- R1.3 Anticon blanket under roof material

GLAZED WINDOWS & DOORS:

- Generic Single Glazed Clear Glass Aluminium Framed (NatHERS Default Windows) Or Glazing Equal To Or Better Than Values Given In This Report

SKYLIGHTS:

- Generic Double Glazed Clear Glass

Disclaimer:

The construction elements/materials summary provided here is for general reference purposes only. In the event of any discrepancies or errors, the details provided in the final NatHERS certificate will supersede/override this summary.

Room schedule

Room	Zone Type	Area (m ²)
GARAGE	Garage	38.29
MASTER BEDROOM	Bedroom	15.40
ENSUITE 2	Night Time	7.87
WALK IN ROBE	Night Time	4.98
WC 1	Day Time	2.53
BED 4	Bedroom	11.40
BED 3	Bedroom	11.40
BED 2	Bedroom	13.09
BATHROOM	Day Time	6.81
WC 2	Unconditioned	2.44
OFFICE/ENTRY	Living	52.33

Room schedule

Room	Zone Type	Area (m ²)
KITCH/DIN/LIV	Kitchen/Living	64.77
PANTRY	Day Time	7.00
RUMPUS	Living	30.86
WIR	Night Time	7.05
BED 1	Bedroom	13.31
ENSUITE 1	Night Time	5.40

Window and glazed door type and performance

Default* windows

Window ID	Window Description	Maximum U-value*	SHGC*	SHGC substitution tolerance ranges	
				lower limit	upper limit
ALM-001-01 A	Aluminium A SG Clear	6.7	0.57	0.54	0.60
ALM-002-01 A	Aluminium B SG Clear	6.7	0.70	0.66	0.73

Custom* windows

Window ID	Window Description	Maximum U-value*	SHGC*	SHGC substitution tolerance ranges	
				lower limit	upper limit

None

Window and glazed door schedule

Location	Window ID	Window no.	Height (mm)	Width (mm)	Window type	Opening %	Orient-ation	Shading device*
BED 1	ALM-001-01 A	W17	1030	2170	Awning	45	WSW	None
BED 2	ALM-001-01 A	W13	600	2170	Awning	45	NNW	None
BED 3	ALM-001-01 A	W12	1030	2170	Awning	45	NNW	None
BED 4	ALM-001-01 A	W11	1030	2170	Awning	45	NNW	None
ENSUITE 2	ALM-001-01 A	W06	1030	1210	Awning	90	SSE	None
GARAGE	ALM-002-01 A	W04	860	2170	Sliding	45	SSE	None
KITCH/DIN/LIV	ALM-002-01 A	W18	600	3576	Fixed	0	NNW	None
KITCH/DIN/LIV	ALM-002-01 A	W07	750	3300	Fixed	0	SSE	None

Window and glazed door *schedule*

Location	Window ID	Window no.	Height (mm)	Width (mm)	Window type	Opening %	Orient-ation	Shading device*
KITCH/DIN/LIV	ALM-002-01 A	W09	2400	3576	Sliding Door	60	WSW	None
KITCH/DIN/LIV	ALM-002-01 A	W10	2400	3576	Sliding Door	60	WSW	None
KITCH/DIN/LIV	ALM-002-01 A	W14	600	3576	Fixed	0	WSW	None
KITCH/DIN/LIV	ALM-002-01 A	W15	600	3576	Fixed	0	WSW	None
MASTER BEDROOM	ALM-001-01 A	W05	1030	2170	Awning	45	SSE	None
OFFICE/ENTRY	ALM-002-01 A	W03	650	1200	Fixed	0	ENE	None
OFFICE/ENTRY	ALM-001-01 A	W01	2400	2840	Awning	45	ENE	None
OFFICE/ENTRY	ALM-002-01 A	W02	1050	2839	Fixed	0	ENE	None
PANTRY	ALM-001-01 A	W08	2100	850	Awning	65	WSW	None
RUMPUS	ALM-001-01 A	W16	1030	2170	Awning	45	WSW	None

Roof window *type and performance value*

Default* roof windows

Window ID	Window Description	Maximum U-value*	SHGC*	SHGC substitution tolerance ranges	
				lower limit	upper limit
None					

Custom* roof windows

Window ID	Window Description	Maximum U-value*	SHGC*	SHGC substitution tolerance ranges	
				lower limit	upper limit
None					

Roof window *schedule*

Location	Window ID	Window no.	Opening %	Height (mm)	Width (mm)	Orient-ation	Outdoor shade	Indoor shade
None								

Skylight *type and performance*

Skylight ID	Skylight description
GEN-04-004a	Double-glazed Clear Skylight

Skylight schedule

Location	Skylight ID	Skylight No.	Skylight shaft length (mm)	Area (m ²)	Orientation	Outdoor shade	Diffuser	Shaft Reflectance
BED 1	GEN-04-004a	SKYLT 03	600	0.77	NNW	None	Yes	80
ENSUITE 1	GEN-04-004a	SKYLT 01	600	0.77	NNW	None	Yes	80
RUMPUS	GEN-04-004a	SKYLT 04	600	0.77	SSE	None	Yes	80
RUMPUS	GEN-04-004a	SKYLT 05	600	0.77	SSE	None	Yes	80
WIR	GEN-04-004a	SKYLT 02	600	0.77	NNW	None	Yes	80

External door schedule

Location	Height (mm)	Width (mm)	Opening %	Orientation
GARAGE	2440	5000	90	ENE
OFFICE/ENTRY	2460	1120	90	ENE

External wall type

Wall ID	Wall Type	Solar absorptance	Wall Colour	Bulk insulation (R-value)	Reflective wall wrap*
BV-NONREFL-CAV-A	Brick Veneer Stud Wall with Non-Reflective Sarking	0.50	Medium	0.00	No
BV-NONREFL-CAV-B	Brick Veneer Stud Wall with Non-Reflective Sarking	0.50	Medium	2.50	No
WB-NONREFL-CAV	Weatherboard Battened (Non-Refl Cavity) Stud Wall	0.50	Medium	2.50	No

External wall schedule

Location	Wall ID	Height (mm)	Width (mm)	Orientation	Horizontal shading feature* projection (mm)	Vertical shading feature
BED 1	WB-NONREFL-CAV	3020	4476	WSW	535	No
BED 2	BV-NONREFL-CAV-B	2600	3706	NNW	1211	Yes
BED 3	BV-NONREFL-CAV-B	2600	3708	NNW	400	Yes
BED 3	BV-NONREFL-CAV-B	2600	811	ENE	3343	Yes
BED 4	BV-NONREFL-CAV-B	2600	3716	NNW	407	Yes
ENSUITE 2	BV-NONREFL-CAV-B	2600	3624	SSE	410	Yes
GARAGE	BV-NONREFL-CAV-A	2600	821	ENE	456	Yes

* Refer to glossary.

External wall *schedule*

Location	Wall ID	Height (mm)	Width (mm)	Orientation	Horizontal shading feature* projection (mm)	Vertical shading feature
GARAGE	BV-NONREFL-CAV-A	2600	6092	SSE	431	Yes
GARAGE	BV-NONREFL-CAV-A	2600	491	WSW	443	Yes
GARAGE	BV-NONREFL-CAV-A	2600	5465	ENE	456	Yes
KITCH/DIN/LIV	BV-NONREFL-CAV-B	3320	5283	NNW	422	Yes
KITCH/DIN/LIV	BV-NONREFL-CAV-B	3320	5283	SSE	420	Yes
KITCH/DIN/LIV	BV-NONREFL-CAV-B	3320	9860	WSW	2989	Yes
MASTER BEDROOM	BV-NONREFL-CAV-B	2600	3505	SSE	409	Yes
OFFICE/ENTRY	BV-NONREFL-CAV-B	3400	1706	SSE	1729	Yes
OFFICE/ENTRY	BV-NONREFL-CAV-B	3100	1949	ENE	2146	Yes
OFFICE/ENTRY	BV-NONREFL-CAV-B	2800	3700	NNW	407	Yes
OFFICE/ENTRY	BV-NONREFL-CAV-B	3465	4451	ENE	440	Yes
OFFICE/ENTRY	BV-NONREFL-CAV-B	3465	804	WSW	3343	Yes
PANTRY	BV-NONREFL-CAV-B	3320	2917	NNW	9841	Yes
PANTRY	BV-NONREFL-CAV-B	3320	2917	SSE		Yes
PANTRY	BV-NONREFL-CAV-B	3320	2401	WSW		Yes
RUMPUS	WB-NONREFL-CAV	3020	3489	WSW	528	No

Internal wall *type*

Wall ID	Wall Type	Area (m ²)	Bulk insulation
INT-PB	Internal Plasterboard Stud Wall	29.8	2.50
INT-PB	Internal Plasterboard Stud Wall	209.5	0.00
INT-PB-EXP1	Internal Plasterboard Stud Wall (exposed 1 side)	53.4	2.50

Floor *type*

Location	Construction	Area (m ²)	Sub-floor ventilation	Added insulation (R-value)	Covering
BATHROOM	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	6.8	N/A	0.62	Tile (10mm)

Floor type

Location	Construction	Area (m ²)	Sub-floor ventilation	Added insulation (R-value)	Covering
BED 1	TIMB-002: Suspended Timber Floor - Lined Below	13.3	N/A	0.15	Tile (10mm)
BED 2	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	13.1	N/A	0.62	Tile (10mm)
BED 3	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	11.4	N/A	0.62	Tile (10mm)
BED 4	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	11.4	N/A	0.62	Tile (10mm)
ENSUITE 1	TIMB-002: Suspended Timber Floor - Lined Below	5.4	N/A	0.15	Tile (10mm)
ENSUITE 2	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	7.9	N/A	0.62	Tile (10mm)
GARAGE	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	38.3	N/A	0.62	Exposed
KITCH/DIN/LIV	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	64.8	N/A	0.62	Tile (10mm)
MASTER BEDROOM	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	15.4	N/A	0.62	Tile (10mm)
OFFICE/ENTRY	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	52.3	N/A	0.62	Tile (10mm)
PANTRY	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	7.0	N/A	0.62	Tile (10mm)
RUMPUS	TIMB-002: Suspended Timber Floor - Lined Below	30.8	N/A	0.15	Tile (10mm)
WALK IN ROBE	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	5.0	N/A	0.62	Tile (10mm)
WC 1	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	2.5	N/A	0.62	Tile (10mm)
WC 2	WAFFLE-85: Concrete Waffle Pod Slab on Ground (85mm)	2.4	N/A	0.62	Tile (10mm)
WIR	TIMB-002: Suspended Timber Floor - Lined Below	7.1	N/A	0.15	Tile (10mm)

Ceiling type

Location	Construction	Bulk insulation (R-value)	Reflective wrap*
BED 1	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
BED 2	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
BED 3	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
BED 4	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
ENSUITE 1	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes

Ceiling type

Location	Construction	Bulk insulation (R-value)	Reflective wrap*
ENSUITE 2	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
GARAGE	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
KITCH/DIN/LIV	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
MASTER BEDROOM	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
OFFICE/ENTRY	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
PANTRY	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
RUMPUS	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes
WIR	ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	4.00	Yes

Ceiling penetrations*

Location	Quantity	Type	Diameter (mm)	Sealed /unsealed
BED 1	2	Downlight	150	Sealed
BED 2	2	Downlight	150	Sealed
BED 3	2	Downlight	150	Sealed
BED 4	2	Downlight	150	Sealed
KITCH/DIN/LIV	11	Downlight	150	Sealed
KITCH/DIN/LIV	1	Exhaust Fan	150	Sealed
MASTER BEDROOM	3	Downlight	150	Sealed
OFFICE/ENTRY	8	Downlight	150	Sealed
PANTRY	1	Downlight	150	Sealed
RUMPUS	5	Downlight	150	Sealed
WALK IN ROBE	1	Downlight	150	Sealed
WIR	1	Downlight	150	Sealed

Ceiling fans

Location	Quantity	Diameter (mm)
BED 1	1	1500



Ceiling fans

Location	Quantity	Diameter (mm)
BED 2	1	1500
BED 3	1	1500
BED 4	1	1500
KITCH/DIN/LIV	2	2100
MASTER BEDROOM	1	1500
OFFICE/ENTRY	1	2100
RUMPUS	1	2100

Roof type

Construction	Added insulation (R-value)	Solar absorptance	Roof Colour
ATTIC-METAL-01: Pitched / Attic Metal Roof (Roofspace) & Flat PB Ceiling	1.30	0.50	Medium

Thermal bridging schedule for steel frame elements

Building element	Steel section dimensions (height x width, mm)	Frame spacing (mm)	Steel thickness (BMT mm)	Thermal Break (R-value)
None				

Appliance schedule

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

Cooling system

Type	Location	Fuel Type	Minimum efficiency / performance	Recommended capacity
No Whole of Home Data				

Heating system

Type	Location	Fuel Type	Minimum efficiency / performance	Recommended capacity
No Whole of Home Data				

Hot water system

Type	Fuel type	Hot Water CER Zone	Minimum efficiency / STC	Assessed daily load [litres]
No Whole of Home Data				



Pool / spa equipment

Type	Fuel type	Minimum efficiency / performance	Recommended capacity
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No Whole of Home Data

Onsite Renewable Energy *schedule*

Type	Orientatation	Generation Capacity [kW]
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No Whole of Home Data

Battery *schedule*

Type	Storage Capacity [kWh]
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No Whole of Home Data

Explanatory Notes

About this report

NatHERS ratings are a reliable guide for comparing different dwelling designs and to demonstrate that designs meet the energy efficiency requirements in the National Construction Code.

NatHERS ratings use computer modelling to evaluate a home's energy efficiency and performance. They use localised climate data and standard assumptions on how people use their home to predict the heating and cooling energy loads and energy value* of the whole home. The thermal performance star rating uses the home's building specifications, layout, orientation and fabric (i.e. walls, windows, floors, roofs and ceilings) to predict the heating and cooling energy loads. The Whole of Home performance rating uses information about the home's appliances and onsite energy generation and storage to estimate the home's energy value*.

The actual energy loads, cost and greenhouse gas emissions of a home may vary from that predicted. This is because the assumptions will not always match the actual occupant usage patterns. For example, the number of occupants and how people use their appliances will vary.

Energy efficient homes use less energy, are warmer on cool days, cooler on hot days and cost less to run.

Accredited assessors

For quality assured NatHERS Certificates, always use an accredited or licenced assessor registered with an Assessor Accrediting Organisation (AAO). AAOs have strict quality assurance processes, and professional development requirements ensuring consistently high standards for assessments.

Non-accredited assessors (Raters) have no ongoing training requirements and

are not quality assured.

Any queries about this report should be directed to the assessor. If the assessor is unable to address questions or concerns, contact the AAO specified on the front of this certificate.

Disclaimer

The NatHERS Certificate format is developed by the NatHERS Administrator. However, the content in the certificate is entered by the assessor. It is the assessor's responsibility to use NatHERS accredited software correctly and follow the NatHERS Technical Note to produce a NatHERS Certificate.

The predicted annual energy load, cost and greenhouse gas emissions in this NatHERS Certificate are an estimate based on an assessment of the dwelling's design by the assessor. It is not a prediction of actual energy use, cost or emissions. The information and ratings may be used to compare how other dwellings are likely to perform when used in a similar way.

Information presented in this report relies on a range of standard assumptions (both embedded in NatHERS accredited software and made by the assessor who prepared this report), including assumptions about occupancy, behaviour, appliance performance, indoor air temperature and local climate.

Not all assumptions made by the assessor using the NatHERS accredited software tool are presented in this report and further details or data files may be obtained from the assessor.

Glossary

Annual energy load	the predicted amount of energy required for heating and cooling, based on standard occupancy assumptions.
AFRC	Australian Fenestration Rating Council
Assessed floor area	the floor area modelled in the software for the purpose of the NatHERS assessment. Note, this may not be consistent with the floor area in the design documents.
Ceiling penetrations	features that require a penetration to the ceiling, including downlights, vents, exhaust fans, range hoods, chimneys and flues. Excludes fixtures attached to the ceiling with small holes through the ceiling for wiring, e.g. ceiling fans; pendant lights, and heating and cooling ducts.
Conditioned	a zone within a dwelling that is expected to require heating and cooling based on standard occupancy assumptions. In some circumstances it will include garages.
COP	Coefficient of performance
Custom windows	windows listed in NatHERS software that are available on the market in Australia and have a WERS (Window Energy Rating Scheme) rating.
Default windows	windows that are representative of a specific type of window product and whose properties have been derived by statistical methods.
EER	Energy Efficiency Ratio, measure of how much cooling can be achieved by an air conditioner for a single kWh of electricity input
Energy use	This is your home's rating without solar or batteries.
Energy value	The net cost to society including, but not limited to, costs to the building user, the environment and energy networks (as defined in the ABCB Housing Provisions Standard).
Entrance door	these signify ventilation benefits in the modelling software and must not be modelled as a door when opening to a minimally ventilated corridor in a Class 2 building.
Exposure	see exposure categories below
Exposure category - exposed	terrain with no obstructions e.g. flat grazing land, ocean-frontage, desert, exposed high-rise unit (usually above 10 floors).
Exposure category - open	terrain with few obstructions at a similar height e.g. grasslands with few well scattered obstructions below 10m, farmland with scattered sheds, lightly vegetated bush blocks, elevated units (e.g. above 3 floors).
Exposure category - suburban	terrain with numerous, closely spaced obstructions below 10m e.g. suburban housing, heavily vegetated bushland areas.
Exposure category - protected	terrain with numerous, closely spaced obstructions over 10 m e.g. city and industrial areas.
Horizontal shading feature	provides shading to the building in the horizontal plane, e.g. eaves, verandahs, pergolas, carports, or overhangs or balconies from upper levels.
National Construction Code (NCC) Class	the NCC groups buildings by their function and use, and assigns a classification code. NatHERS software models NCC Class 1, 2 or 4 buildings and attached Class 10a buildings. Definitions can be found at www.abcb.gov.au .
Net zero home	a home that achieves a net zero energy value*.
Opening percentage	the openability percentage or operable (moveable) area of doors or windows that is used in ventilation calculations.
Provisional value	an assumed value that does not represent an actual value. For example, if the wall colour is unspecified in the documentation, a provisional value of 'medium' must be modelled. Acceptable provisional values are outlined in the NatHERS Technical Note and can be found at www.nathers.gov.au
Recommended capacity	this is the capacity or size of equipment that is recommended by NatHERS to achieve the desired comfort conditions in the zone or zones serviced. This is a recommendation and the final selection sizing should be confirmed by a suitably qualified person.
Reflective wrap (also known as foil)	can be applied to walls, roofs and ceilings. When combined with an appropriate airgap and emissivity value, it provides insulative properties.
Roof window	for NatHERS this is typically an operable window (i.e. can be opened), will have a plaster or similar light well if there is an attic space, and generally does not have a diffuser.
Shading features	includes neighbouring buildings, fences, and wing walls, but excludes eaves.
Solar heat gain coefficient (SHGC)	the fraction of incident solar radiation admitted through a window, both directly transmitted as well as absorbed and subsequently released inward. SHGC is expressed as a number between 0 and 1. The lower a window's SHGC, the less solar heat it transmits.
Skylight (also known as roof lights)	for NatHERS this is typically a moulded unit with flexible reflective tubing (light well) and a diffuser at ceiling level.
STCs	Small-scale Technology Certificates, certificates created by the REC registry for renewable energy technologies that may be bought and sold as part of the Small-scale Renewable Energy Scheme operated by the Clean Energy Regulatory
Thermal breaks	are materials with an R-value greater than or equal to 0.2 that must separate the metal frame from the cladding. This includes, but is not limited to, materials such as timber battens greater than or equal to 20mm thick, continuous thermal breaks such as polystyrene insulation sheeting, plastic strips or furring channels.
U-value	the rate of heat transfer through a window. The lower the U-value, the better the insulating ability.
Unconditioned	a zone within a dwelling that is assumed to not require heating and cooling based on standard occupancy assumptions
Vertical shading features	provides shading to the building in the vertical plane and can be parallel or perpendicular to the subject wall/window. Includes privacy screens, other walls in the building (wing walls), fences, other buildings, vegetation (protected or listed heritage trees).
Window shading device	a device fixed to windows that provides shading e.g. window awnings or screens but excludes horizontal* or vertical shading features* (eg eaves and balconies)

* Refer to glossary.